Generative Models in Finance

Week 4: Flow Matching and Diffusion Models — Foundations

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Overview

Reference: P. Holderrieth & E. Erives, *An Introduction to Flow Matching and Diffusion Models*, MIT 6.S184, arXiv:2506.02070, 2025.

Part 1: Generative Modelling as Sampling

- Objects to generate (images, videos, molecules) are vectors $z \in \mathbb{R}^d$
- The diversity of valid objects is modelled by a data distribution p_{data} over $\mathbb{R}^d$
- Generation = Sampling: generating z means sampling $z \sim p_{\mathrm{data}}$
- Dataset: i.i.d. samples $z_1, \ldots, z_N \sim p_{\mathrm{data}}$
- Conditional generation: sample $z \sim p_{\mathrm{data}}(\cdot \mid y)$ for a conditioning variable y (e.g. text prompt). We focus on unconditional generation first
- Goal: Learn to transform samples from $p_{\mathrm{init}} = \mathcal{N}(0, I_d)$ into samples from p_{data}, via simulation of a suitably constructed *differential equation*

Part 2: Flow Models — ODEs

- A trajectory is a function $X : [0,1] \to \mathbb{R}^d$, $t \mapsto X_t$
- A vector field is a function

$$u : \mathbb{R}^d \times [0,1] \to \mathbb{R}^d, \quad (x,t) \mapsto u_t(x),$$

  specifying a velocity at every point x and time t
- An ordinary differential equation (ODE) with initial condition is:

$$\frac{\mathrm{d}}{\mathrm{d}t} X_t = u_t(X_t), \tag{4.1}$$

$$X_0 = x_0. \tag{4.2}$$

- Equation (4.1) says: the instantaneous rate of change of X_t is prescribed by the vector field u evaluated at the current position X_t

Flows

- The flow of the ODE is the function

$$\psi : \mathbb{R}^d \times [0,1] \to \mathbb{R}^d, \quad (x_0, t) \mapsto \psi_t(x_0)$$

  that maps each initial condition x_0 to the solution $X_t = \psi_t(x_0)$ at time t
- The flow satisfies:

$$\frac{\mathrm{d}}{\mathrm{d}t}\psi_t(x_0) = u_t(\psi_t(x_0)), \tag{4.3}$$

$$\psi_0(x_0) = x_0. \tag{4.4}$$

- Intuition: Vector fields, ODEs, and flows are three descriptions of the same object: *vector fields define ODEs whose solutions are flows*

Existence and Uniqueness

Theorem 4.1 (Flow existence and uniqueness)

If $u : \mathbb{R}^d \times [0, 1] \to \mathbb{R}^d$ is continuously differentiable with bounded derivative, then the ODE (4.1)–(4.2) has a unique solution given by a flow ψ_t. Moreover, ψ_t is a diffeomorphism for all t, i.e. ψ_t is continuously differentiable with continuously differentiable inverse ψ_t^{-1}.

- Since we parameterise $u_t(x)$ with neural networks with smooth activations (e.g. GELU, SiLU), the assumptions are approximately satisfied in practice
- Takeaway: Flows exist, are unique, and are diffeomorphisms — they "warp" space in an invertible way

Example: Linear Vector Fields

Example 4.2 (Linear vector field)

Let $u_t(x) = -\theta x$ *for* $\theta > 0$*. Then the flow is*

$$\psi_t(x_0) = \exp(-\theta t)\, x_0. \tag{4.5}$$

- Verification: $\psi_0(x_0) = x_0$ ✓, and

$$\frac{\mathrm{d}}{\mathrm{d}t}\psi_t(x_0) = -\theta \exp(-\theta t)\, x_0 = -\theta\psi_t(x_0) = u_t(\psi_t(x_0)) \quad \checkmark$$

- This flow contracts all points exponentially towards the origin

Simulating an ODE: ODE Solvers

- In general, ψ_t cannot be computed in closed form; we approximate it numerically
- Write $t_k = kh$ for $k = 0, \ldots, n$ with step size $h = n^{-1}$, and let $X_{t_0} = x_0$
- Euler method (order 1): one evaluation of u per step

$$X_{t_{k+1}} = X_{t_k} + h\, u_{t_k}(X_{t_k}) \tag{4.6}$$

- The order of a solver measures accuracy: an order-p method has global error $\mathcal{O}(h^p)$
- Higher-order methods use *multiple evaluations* of u per step to achieve better accuracy for the same step size

ODE Solvers: Higher-Order Methods

- Midpoint method (order 2): evaluate u at the midpoint

$$\begin{aligned} X_{t_k+h/2} &= X_{t_k} + \tfrac{h}{2}\, u_{t_k}(X_{t_k}) \\ X_{t_{k+1}} &= X_{t_k} + h\, u_{t_k+h/2}(X_{t_k+h/2}) \end{aligned}$$

  - half-step
  - full step

- Heun's method (order 2): predictor-corrector strategy

$$\begin{aligned} \tilde{X}_{t_{k+1}} &= X_{t_k} + h\, u_{t_k}(X_{t_k}) \\ X_{t_{k+1}} &= X_{t_k} + \frac{h}{2}\big(u_{t_k}(X_{t_k}) + u_{t_{k+1}}(\tilde{X}_{t_{k+1}})\big) \end{aligned}$$

  - predictor (Euler)
  - corrector

- Both use 2 evaluations of u per step; same order, different geometric properties

ODE Solvers: Runge–Kutta Methods

- The classical 4th-order Runge–Kutta method (RK4):

$$\begin{aligned} k_1 &= u_{t_k}(X_{t_k}), \\ k_2 &= u_{t_k+h/2}\big(X_{t_k} + \tfrac{h}{2}\, k_1\big), \\ k_3 &= u_{t_k+h/2}\big(X_{t_k} + \tfrac{h}{2}\, k_2\big), \\ k_4 &= u_{t_{k+1}}\big(X_{t_k} + h\, k_3\big), \end{aligned}$$

$$X_{t_{k+1}} = X_{t_k} + \frac{h}{6}\big(k_1 + 2k_2 + 2k_3 + k_4\big)$$

- Uses 4 evaluations of u per step; global error $\mathcal{O}(h^4)$
- Euler, midpoint, and Heun are all special cases of the general family of Runge–Kutta methods, characterised by their Butcher tableau

ODE Solvers: Practical Considerations

- Cost–accuracy trade-off: each step of an order-p method requires $s \geq p$ evaluations of u, but fewer steps are needed for the same accuracy
- Adaptive step-size control: modern solvers (e.g. Dormand–Prince / `dopri5`) *adapt* h at each step by comparing two estimates of different orders
- In practice: Euler is often sufficient for flow/diffusion model sampling because many steps are cheap; Heun/midpoint offer easy improvements; RK4 or adaptive solvers are used when high accuracy is critical
- For the rest of these lectures, we use the Euler method for simplicity, but all results generalise to higher-order solvers

Flow Models

- A flow model is defined by:

$$X_0 \sim p_{\mathrm{init}}$$
▶ random initialisation

$$\frac{\mathrm{d}}{\mathrm{d}t} X_t = u_t^\theta(X_t)$$
▶ ODE

where $u_t^\theta : \mathbb{R}^d \times [0,1] \to \mathbb{R}^d$ is a neural network with parameters θ

- Goal: Choose θ so that the endpoint X_1 has distribution p_{data}, i.e.

$$X_1 \sim p_{\mathrm{data}} \quad \Leftrightarrow \quad \psi_1^\theta(X_0) \sim p_{\mathrm{data}}$$

- Important: The neural network parameterises the *vector field*, not the flow itself. The flow is obtained by simulating the ODE

Sampling from a Flow Model

Algorithm 1: Sampling from a Flow Model (Euler method)

Require: Neural network vector field u_t^θ, number of steps n

1. Set $t = 0$, step size $h = 1/n$
2. Draw $X_0 \sim p_{\mathrm{init}}$
3. For $i = 1, \ldots, n$ do:
4. $\quad X_{t+h} = X_t + h\, u_t^\theta(X_t)$
5. $\quad t \leftarrow t + h$
6. Return X_1

Part 3: Diffusion Models — Brownian Motion

Definition 4.3 (Brownian motion)

A Brownian motion *(or* Wiener process*)* $W = (W_t)_{0 \le t \le 1}$ *is a stochastic process such that* $W_0 = 0$*, the trajectories* $t \mapsto W_t$ *are continuous, and:*

1. *Normal increments:* $W_t - W_s \sim \mathcal{N}(0, (t-s)I_d)$ *for all* $0 \le s < t$
2. *Independent increments: For any* $0 \le t_0 < t_1 < \cdots < t_n$*, the increments* $W_{t_1} - W_{t_0}, \ldots, W_{t_n} - W_{t_{n-1}}$ *are independent*

- Approximate simulation: set $W_0 = 0$ and update

$$W_{t+h} = W_t + \sqrt{h}\,\epsilon_t, \quad \epsilon_t \sim \mathcal{N}(0, I_d) \tag{4.7}$$

- Brownian paths are continuous but nowhere differentiable (a.s.)
- Brownian motion is the fundamental building block of stochastic calculus, analogous to the Gaussian distribution in probability theory

Advanced BM Simulation: Brownian Bridge and Virtual Brownian Tree

- The sequential scheme (4.7) generates W on a *fixed* grid; adaptive solvers may query W at *arbitrary* times
- Brownian bridge interpolation: Given W_s and W_t, for any $s < r < t$:

$$W_r \mid W_s, W_t \ \sim \ \mathcal{N}\Big(\frac{(t-r)W_s + (r-s)W_t}{t-s},\ \frac{(r-s)(t-r)}{t-s}\, I_d\Big)$$

  This allows generating BM values at new times *conditionally* on already-sampled values
- Virtual Brownian Tree[1]: organise time intervals in a *binary tree*; at each query time r, walk down the tree and apply Brownian bridge interpolation
- The entire path is determined by a single PRNG seed: previously generated values need not be stored $\Rightarrow \mathcal{O}(1)$ memory, $\mathcal{O}(\log(1/\mathrm{tol}))$ time per query
- This enables *adaptive step-size* SDE solvers: the solver chooses h on the fly while maintaining a consistent Brownian path across rejected/accepted steps
- Implementation available in Diffrax[2], a JAX library for numerical differential equation solvers

[1] Kidger, *On Neural Differential Equations*, PhD thesis, Oxford, 2021. Extended by Jelinčič, Foster & Kidger, *Single-seed generation of Brownian paths and integrals for adaptive and high order SDE solvers*, arXiv:2405.06464, 2024.

[2] `https://github.com/patrick-kidger/diffrax`

From ODEs to SDEs

- An ODE $\mathrm{d}X_t = u_t(X_t)\,\mathrm{d}t$ defines a *deterministic* flow; given $X_0 = x_0$, the trajectory is uniquely determined
- An Itô stochastic differential equation perturbs this with a martingale driving term:

$$\mathrm{d}X_t = u_t(X_t)\,\mathrm{d}t + \sigma_t\,\mathrm{d}W_t, \tag{4.8}$$

$$X_0 = x_0, \tag{4.9}$$

  where $\sigma_t \geq 0$ is the diffusion coefficient and W is a d-dimensional Brownian motion
- The integral form makes the meaning precise: for all $0 \leq s < t \leq 1$,

$$X_t = X_s + \int_s^t u_r(X_r)\,\mathrm{d}r + \int_s^t \sigma_r\,\mathrm{d}W_r \tag{4.10}$$

  where the second integral is an Itô integral (defined as an L^2-limit of Riemann sums against W)
- Key distinction from ODEs: The solution (X_t) is now a *stochastic process*, not a deterministic path. There is no flow map ψ_t; instead, X_t defines a random variable for each t, adapted to the filtration generated by W

SDE Existence and Uniqueness

Theorem 4.4 (Strong existence and pathwise uniqueness[a])

[a] Mao, X. *Stochastic Differential Equations and Applications*, 2nd ed., Horwood, 2007, Ch. 3.

Suppose that:

1. *(Lipschitz continuity) There exists $L > 0$ such that for all $x, y \in \mathbb{R}^d$ and $t \in [0,1]$:*
$$\|u_t(x) - u_t(y)\| \leq L\|x - y\|$$
2. *(Linear growth) There exists $K > 0$ such that for all $x \in \mathbb{R}^d$ and $t \in [0,1]$:*
$$\|u_t(x)\|^2 + |\sigma_t|^2 \leq K^2(1 + \|x\|^2)$$

Then the SDE (4.8)–(4.9) admits a unique strong solution: a continuous, $(\mathcal{F}_t^W)$-adapted process $(X_t)_{0 \leq t \leq 1}$ satisfying (4.10) a.s., with $\mathbb{E}[\sup_{0 \leq t \leq 1} \|X_t\|^2] < \infty$.

- The proof proceeds by Picard iteration on the integral equation (4.10), using the Itô isometry and the Borel–Cantelli lemma to upgrade L^2-convergence to a.s. uniform convergence
- Every ODE is a special case with $\sigma_t = 0$. Henceforth, when we speak about SDEs, we consider ODEs as a special case

The Ornstein–Uhlenbeck Process

Example 4.5 (Ornstein–Uhlenbeck process)

Let $u_t(x) = -\theta x$ for $\theta > 0$ and $\sigma_t = \sigma \geq 0$ constant. The SDE

$$\mathrm{d}X_t = -\theta X_t \,\mathrm{d}t + \sigma \,\mathrm{d}W_t \tag{4.11}$$

defines an Ornstein–Uhlenbeck (OU) process.

- The drift is globally Lipschitz with $L = \theta$, so the theorem applies
- Closed-form solution (by variation of constants / integrating factor $e^{\theta t}$):

$$X_t = e^{-\theta t} x_0 + \sigma \int_0^t e^{-\theta(t-s)} \,\mathrm{d}W_s$$

- Since the Itô integral of a deterministic integrand is Gaussian:

$$X_t \sim \mathcal{N}\left(e^{-\theta t} x_0,\ \frac{\sigma^2}{2\theta}(1 - e^{-2\theta t})\, I_d\right)$$

- As $t \to \infty$: $X_t \xrightarrow{d} \mathcal{N}(0, \frac{\sigma^2}{2\theta} I_d)$ (stationary distribution)
- For $\sigma = 0$: reduces to the deterministic flow $\psi_t(x_0) = e^{-\theta t} x_0$ from (4.5)

SDE Solvers: Euler–Maruyama Method

- The Euler–Maruyama method[3] is the SDE analogue of the Euler method for ODEs:

$$X_{t_{k+1}} = X_{t_k} + h\, u_{t_k}(X_{t_k}) + \sigma_{t_k}(W_{t_{k+1}} - W_{t_k}) \tag{4.12}$$

  Since $W_{t_{k+1}} - W_{t_k} \sim \mathcal{N}(0, h\, I_d)$, equivalently: $X_{t_{k+1}} = X_{t_k} + h\, u_{t_k}(X_{t_k}) + \sqrt{h}\, \sigma_{t_k}\, \epsilon_k$, $\epsilon_k \sim \mathcal{N}(0, I_d)$
- When $\sigma_t = 0$, this reduces to the Euler method (4.6)
- Two notions of convergence for SDE solvers:
  - Strong order p: $\mathbb{E}[\|X_T - X_T^h\|] = \mathcal{O}(h^p)$ (pathwise accuracy)
  - Weak order q: $|\mathbb{E}[f(X_T)] - \mathbb{E}[f(X_T^h)]| = \mathcal{O}(h^q)$ for smooth f (distributional accuracy)
- Euler–Maruyama has strong order $1/2$ and weak order 1

[3] Maruyama, G. *Continuous Markov processes and stochastic equations*, Rend. Circ. Mat. Palermo, 4:48–90, 1955.

SDE Solvers: Milstein Method

- Euler–Maruyama ignores the interaction between noise and drift. The Milstein method[4] adds a correction term from the Itô–Taylor expansion:

$$\begin{aligned} X_{t_{k+1}} = X_{t_k} &+ h\, u_{t_k}(X_{t_k}) + \sigma_{t_k}\, \Delta W_k \\ &+ \tfrac{1}{2}\, \sigma_{t_k}\, \sigma'_{t_k} \left((\Delta W_k)^2 - h\, I_d \right) \end{aligned}$$

  where $\Delta W_k = W_{t_{k+1}} - W_{t_k}$ and $\sigma'_{t_k} = \frac{\mathrm{d}\sigma}{\mathrm{d}t}\big|_{t_k}$

- This achieves strong order 1 (and weak order 1), matching the ODE Euler method's convergence rate
- Caveat (multidimensional case): When $d > 1$ and the diffusion coefficient depends on x, the Milstein correction requires simulating Lévy areas $\int_{t_k}^{t_{k+1}} (W_s^i - W_{t_k}^i)\, \mathrm{d}W_s^j$, which is non-trivial
- For scalar noise ($d = 1$) or additive noise (σ_t independent of x), Milstein is straightforward and recommended over Euler–Maruyama

[4] Milstein, G.N. *Approximate integration of stochastic differential equations*, Theory Probab. Appl., 19(3):557–562, 1974.

SDE Solvers: Higher-Order and Adaptive Methods

- Stochastic Runge–Kutta methods[5]: Achieve strong order 1 (or higher) *without* explicit derivatives of u or σ, by evaluating the coefficients at multiple intermediate points — analogous to deterministic RK methods
- Key difficulty vs. ODEs: Achieving strong order > 1 requires simulating iterated stochastic integrals (Lévy areas), which adds computational cost; most practical SDE solvers therefore operate at strong order ≤ 1
- Adaptive step-size control: Possible via the Virtual Brownian Tree (cf. earlier slide); reject steps where the local error estimate exceeds a tolerance, and retry with smaller h
- For *weak* approximation (sufficient for sampling in generative models), higher weak-order methods exist that avoid Lévy areas entirely

General reference: Kloeden, P.E. & Platen, E. *Numerical Solution of Stochastic Differential Equations*, Springer, 1992.

[5] Rößler, A. *Runge–Kutta methods for the strong approximation of solutions of SDEs*, SIAM J. Numer. Anal., 48(3):922–952, 2010.

The Rough Path Perspective on SDEs

- Core problem: The Itô integral $\int_s^t f(X_r)\,\mathrm{d}W_r$ is defined via L^2-limits. This *probabilistic* construction obscures the *analytic* structure needed for high-order numerics
- Lyons' rough paths theory[6]: Reformulate the SDE as a *pathwise* differential equation driven by the Brownian motion viewed as a rough path
- A rough path over W is not just the path $t \mapsto W_t$, but an enhanced path that also encodes iterated integrals:

$$\mathbf{W}_t = \left(W_t,\ \mathbb{W}_{s,t} := \int_s^t (W_r - W_s) \otimes \circ \mathrm{d}W_r \right)$$

  The second component $\mathbb{W}_{s,t} \in \mathbb{R}^{d\times d}$ is the Lévy area (the antisymmetric part encodes the "winding" of the path). The integral is in the Stratonovich sense
- Key insight: Once $\mathbf{W}$ is fixed, the solution map $\mathbf{W} \mapsto X$ is *continuous* in rough path topology — the Universal Limit Theorem[7]. This yields solutions in the Stratonovich sense; the Itô solution is recovered via the Itô–Stratonovich correction:

$$\int_s^t f(X_r) \circ \mathrm{d}W_r = \int_s^t f(X_r)\,\mathrm{d}W_r + \tfrac{1}{2} \int_s^t f'(X_r)\,\sigma_r\,\mathrm{d}r$$

[6] Lyons, T.J. *Differential equations driven by rough signals*, Rev. Mat. Iberoam., 14(2):215–310, 1998.

[7] Friz, P.K. & Hairer, M. *A Course on Rough Paths*, 2nd ed., Springer, 2020, Ch. 8.

Rough Paths and Numerical Schemes

- The ULT implies: to approximate X, it suffices to approximate the *driving rough path* **W** well. Since the ULT operates in the Stratonovich setting, the numerical schemes below approximate the Stratonovich SDE

$$\circ \mathrm{d}X_t = \tilde{u}_t(X_t)\,\mathrm{d}t + \sigma_t \circ \mathrm{d}W_t$$

  where $\tilde{u}_t(x) = u_t(x) - \frac{1}{2}\sigma_t\sigma_t'$ absorbs the Itô–Stratonovich drift correction
- Davie's scheme[8]: Given the Brownian increment $\Delta W_k = W_{t_{k+1}} - W_{t_k}$ and the Lévy area $\mathbb{W}_{t_k,t_{k+1}}$, update via

$$X_{t_{k+1}} = X_{t_k} + \tilde{u}_{t_k}(X_{t_k})\,h + \sigma_{t_k}\,\Delta W_k + D\sigma_{t_k}(X_{t_k})\,\sigma_{t_k}\,\mathbb{W}_{t_k,t_{k+1}}$$

- This achieves strong order 1 and makes explicit *why* Lévy areas are needed: they are part of the rough path data at the required level of regularity
- The Milstein scheme (in Stratonovich form) is recovered as a special case

[8] Davie, A.M. *Differential equations driven by rough paths: an approach via discrete approximation*, Appl. Math. Res. Express, 2008, arXiv:0710.0772.

Log-ODE Method and Higher-Order Rough Path Solvers

- Log-ODE method[9]: On each interval $[t_k, t_{k+1}]$, replace the rough driving signal by a *smooth* path (a geodesic in the free nilpotent group) that matches the same iterated integrals, then solve the resulting ODE exactly or to high order
- This transforms the SDE solve into a sequence of *local ODE solves*, for which classical Runge–Kutta methods apply
- Convergence: Approximating the rough path signature to level N yields a method of strong order $N/2 - \varepsilon$. Level $N = 2$ (increments + Lévy area) gives strong order 1; level $N = 3$ adds further iterated integrals for strong order $3/2$
- Dimension-free error bounds[10]: Rough path estimates give error bounds that depend on the *regularity* of the driving signal, not the ambient dimension d — a crucial advantage for high-dimensional problems in generative modelling

General reference: Friz, P.K. & Victoir, N.B. *Multidimensional Stochastic Processes as Rough Paths*, Cambridge, 2010.

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[9] Foster, J., Lyons, T. & Oberhauser, H. *An optimal polynomial approximation of Brownian motion*, SIAM J. Numer. Anal., 58(3):1393–1421, 2020, arXiv:1904.06998.

[10] Boutaib, Y., Gyurkó, L.G., Lyons, T. & Yang, D. *Dimension-free Euler estimates of rough differential equations*, Rev. Roumaine Math. Pures Appl., 59(1):25–53, 2014, arXiv:1307.4708.

Diffusion Models

- A diffusion model is defined by:

$$\mathrm{d}X_t = u_t^\theta(X_t)\,\mathrm{d}t + \sigma_t\,\mathrm{d}W_t \qquad \blacktriangleright \text{ SDE}$$

$$X_0 \sim p_{\mathrm{init}} \qquad \blacktriangleright \text{ random initialisation}$$

- It consists of:
  - A neural network $u^\theta : \mathbb{R}^d \times [0,1] \to \mathbb{R}^d$, $(x,t) \mapsto u_t^\theta(x)$ with parameters θ
  - A fixed diffusion coefficient $\sigma_t : [0,1] \to [0,\infty)$
- Goal: $X_1 \sim p_{\mathrm{data}}$
- A diffusion model with $\sigma_t = 0$ is a flow model

Sampling from a Diffusion Model

Algorithm 2: Sampling from a Diffusion Model (Euler–Maruyama)

Require: Neural network u_t^θ, number of steps n, diffusion coefficient σ_t

1. Set $t = 0$, step size $h = 1/n$
2. Draw $X_0 \sim p_{\mathrm{init}}$
3. For $i = 1, \ldots, n$ do:
4. Draw $\epsilon \sim \mathcal{N}(0, I_d)$
5. $X_{t+h} = X_t + h\, u_t^\theta(X_t) + \sigma_t \sqrt{h}\, \epsilon$
6. $t \leftarrow t + h$
7. Return X_1

Part 4: Constructing the Training Target

- If we randomly initialise θ, simulating the ODE/SDE produces nonsense
- We need to *train* the neural network by minimising a loss:

$$\mathcal{L}(\theta) = \|u_t^\theta(x) - u_t^{\mathrm{target}}(x)\|^2 \tag{4.13}$$

- The training target $u_t^{\mathrm{target}} : \mathbb{R}^d \times [0,1] \to \mathbb{R}^d$ is a vector field whose corresponding ODE/SDE converts p_{init} into p_{data}
- This part: Derive a formula for u_t^{target}
- Next part: Design a tractable training algorithm to approximate u_t^{target}

Conditional Probability Paths

Definition 4.6 (Conditional probability path)

A conditional (interpolating) probability path is a family of distributions $p_t(\cdot \mid z)$ over $\mathbb{R}^d$, parameterised by $t \in [0, 1]$ and $z \in \mathbb{R}^d$, such that

$$p_0(\cdot \mid z) = p_{\mathrm{init}}, \qquad p_1(\cdot \mid z) = \delta_z, \qquad \textit{for all } z \in \mathbb{R}^d. \tag{4.14}$$

- Here δ_z is the Dirac delta at z (the distribution that always returns z)
- Intuitively, $p_t(\cdot \mid z)$ gradually converts a *single* data point z from noise p_{init} at $t = 0$ to a point mass at z at $t = 1$
- Think of this as a trajectory in the space of distributions, indexed by t

Marginal Probability Paths

Definition 4.7 (Marginal probability path)

Given a conditional probability path $\{p_t(\cdot \mid z)\}_{t\in[0,1],z\in\mathbb{R}^d}$ *and a data distribution* p_{data}*, the* *marginal probability path* $\{p_t\}_{t\in[0,1]}$ *is the mixture*

$$p_t(x) = \int_{\mathbb{R}^d} p_t(x \mid z)\, p_{\mathrm{data}}(z)\,\mathrm{d}z = \mathbb{E}_{z\sim p_{\mathrm{data}}}\big[p_t(x \mid z)\big]. \tag{4.15}$$

- Equivalent sampling procedure:

$$z \sim p_{\mathrm{data}}, \quad x \sim p_t(\cdot \mid z) \quad \Rightarrow \quad x \sim p_t \tag{4.16}$$

- Boundary conditions (inherited from the conditional path):

$$p_0(x) = \int p_0(x \mid z)\, p_{\mathrm{data}}(z)\,\mathrm{d}z = \int p_{\mathrm{init}}(x)\, p_{\mathrm{data}}(z)\,\mathrm{d}z = p_{\mathrm{init}}(x), \tag{4.17}$$

$$p_1(x) = \int \delta_z(x)\, p_{\mathrm{data}}(z)\,\mathrm{d}z = p_{\mathrm{data}}(x). \tag{4.18}$$

- Computational note: We can *sample* from p_t via (4.16), but we cannot *evaluate* $p_t(x)$ because the integral (4.15) is intractable (it marginalises over the entire data distribution)

Gaussian Conditional Probability Paths

Example 4.8 (Gaussian conditional probability path)

Let α_t, β_t be noise schedulers: continuously differentiable, monotonic functions with $\alpha_0 = \beta_1 = 0$ and $\alpha_1 = \beta_0 = 1$. Define

$$p_t(\cdot \mid z) = \mathcal{N}(\alpha_t z, \beta_t^2 I_d). \tag{4.19}$$

- Check: $p_0(\cdot \mid z) = \mathcal{N}(0, \beta_0^2 I_d) = \mathcal{N}(0, I_d) = p_{\text{init}}$ ✓
- Check: $p_1(\cdot \mid z) = \mathcal{N}(\alpha_1 z, 0) = \delta_z$ ✓
- Sampling from the marginal path:

$$z \sim p_{\text{data}}, \quad \epsilon \sim \mathcal{N}(0, I_d) \quad \Rightarrow \quad x = \alpha_t z + \beta_t \epsilon \sim p_t \tag{4.20}$$

- Intuitively: more noise for smaller t (closer to $t = 0$, pure noise), less noise for larger t (closer to $t = 1$, pure data)

Interlude: The Wasserstein-2 Distance

- Optimal transport[11] asks: what is the cheapest way to move mass from a distribution μ to a distribution ν, when transporting a unit of mass from x to y costs $\|x-y\|^2$?
- The Wasserstein-2 distance solves this by optimising over all couplings (joint distributions with the correct marginals):

$$W_2^2(\mu,\nu) = \inf_{\gamma\in\Pi(\mu,\nu)} \int_{\mathbb{R}^d\times\mathbb{R}^d} \|x-y\|^2 \,\mathrm{d}\gamma(x,y),$$

  where $\Pi(\mu,\nu) = \{\gamma \in \mathcal{P}(\mathbb{R}^d\times\mathbb{R}^d) : \gamma(\cdot,\mathbb{R}^d)=\mu,\ \gamma(\mathbb{R}^d,\cdot)=\nu\}$
- Geodesic: $(\mathcal{P}_2(\mathbb{R}^d), W_2)$ is a metric space[12]. A constant-speed geodesic is a curve $(\mu_t)_{t\in[0,1]}$ satisfying

$$W_2(\mu_s,\,\mu_t) = |t-s|\, W_2(\mu_0,\,\mu_1), \qquad \forall\, s,t\in[0,1].$$

- Displacement interpolation[13]: given an optimal transport map T, the path $\mu_t = \big((1-t)\,\mathrm{id} + t\,T\big)_{\#}\mu$ is the unique constant-speed geodesic from $\mu_0=\mu$ to $\mu_1 = T_{\#}\mu = \nu$

[11] Villani, C. (2009). *Optimal Transport: Old and New*. Springer. Grundlehren der mathematischen Wissenschaften, Vol. 338.

[12] Ambrosio, L., Gigli, N., Savaré, G. (2008). *Gradient Flows in Metric Spaces and in the Space of Probability Measures*. 2nd ed., Birkhäuser. Lectures in Mathematics ETH Zürich.

[13] McCann, R.J. (1997). A convexity principle for interacting gases. *Advances in Mathematics*, 128(1), 153–179.

The Conditional Optimal Transport Path

Set $\alpha_t = t$, $\beta_t = 1 - t$[14]:

$$p_t(\cdot \mid z) = \mathcal{N}\big(t\, z,\ (1-t)^2 I_d\big), \qquad X_t = t\, z + (1-t)\, \epsilon, \quad \epsilon \sim \mathcal{N}(0, I_d). \quad (4.21)$$

- The unique OT map from $\mathcal{N}(0, I_d)$ to δ_z is $T(x) = z$ (all mass goes to z)
- The displacement interpolation $T_t(x) = (1-t)\, x + t\, z$ recovers exactly (4.21)
- Hence the CondOT path is the W_2-*geodesic* from $\mathcal{N}(0, I_d)$ to δ_z
- Key consequence: Each sample follows a *straight line* with constant velocity:

$$\dot{X}_t = z - \epsilon$$

Straighter trajectories $\Rightarrow$ fewer ODE discretisation steps at inference

[14] Lipman, Y., Chen, R.T.Q., Ben-Hamu, H., Nickel, M. (2023). Flow Matching for Generative Modeling. *ICLR 2023*. arXiv:2210.02747.

Conditional Vector Fields

- We now construct a vector field whose ODE trajectories follow the conditional probability path
- A conditional vector field $u_t^{\mathrm{target}}(\cdot \mid z)$ is defined so that its ODE yields $X_t \sim p_t(\cdot \mid z)$:

$$X_0 \sim p_{\mathrm{init}}, \quad \frac{\mathrm{d}}{\mathrm{d}t} X_t = u_t^{\mathrm{target}}(X_t \mid z) \quad \Rightarrow \quad X_t \sim p_t(\cdot \mid z). \tag{4.22}$$

Conditional Gaussian Vector Field

Example 4.9 (Target ODE for Gaussian probability paths)

For $p_t(\cdot \mid z) = \mathcal{N}(\alpha_t z, \beta_t^2 I_d)$, the conditional Gaussian vector field is

$$u_t^{\text{target}}(x \mid z) = \left(\dot{\alpha}_t - \frac{\dot{\beta}_t}{\beta_t}\alpha_t\right) z + \frac{\dot{\beta}_t}{\beta_t} x. \tag{4.23}$$

Proof. Define the map $\psi_t^{\text{target}}(x \mid z) = \alpha_t z + \beta_t x$. This is a valid flow:

- $\psi_0^{\text{target}}(x \mid z) = \alpha_0 z + \beta_0 x = 0 + 1 \cdot x = x \quad (\psi_0 = \mathrm{id})$ ✓
- For $t \in [0, 1)$, $\beta_t > 0$, so $\psi_t^{\text{target}}(\cdot \mid z)$ is a diffeomorphism with inverse $(\psi_t^{\text{target}})^{-1}(y \mid z) = (y - \alpha_t z)/\beta_t$ ✓

If $X_0 \sim \mathcal{N}(0, I_d) = p_{\text{init}}$, then by the affine property of Gaussians:

$$X_t = \psi_t^{\text{target}}(X_0 \mid z) = \alpha_t z + \beta_t X_0 \sim \mathcal{N}(\alpha_t z, \beta_t^2 I_d) = p_t(\cdot \mid z). \tag{4.24}$$

Conditional Gaussian Vector Field (Proof cont.)

It remains to extract $u_t^{\mathrm{target}}(x \mid z)$ from ψ_t^{target}. By definition of a flow (4.3):

$$\frac{\mathrm{d}}{\mathrm{d}t}\psi_t^{\mathrm{target}}(x \mid z) = u_t^{\mathrm{target}}(\psi_t^{\mathrm{target}}(x \mid z) \mid z) \quad \text{for all } x, z \in \mathbb{R}^d.$$

Differentiating $\psi_t^{\mathrm{target}}(x \mid z) = \alpha_t z + \beta_t x$ with respect to t:

$$\dot{\alpha}_t z + \dot{\beta}_t x = u_t^{\mathrm{target}}(\alpha_t z + \beta_t x \mid z).$$

Since $\beta_t > 0$ for $t \in [0, 1)$, the map $x \mapsto \alpha_t z + \beta_t x$ is surjective onto $\mathbb{R}^d$.
Substituting $x = (\psi_t^{\mathrm{target}})^{-1}(y \mid z) = (y - \alpha_t z)/\beta_t$:

$$\dot{\alpha}_t z + \dot{\beta}_t \cdot \frac{y - \alpha_t z}{\beta_t} = u_t^{\mathrm{target}}(y \mid z) \quad \text{for all } y \in \mathbb{R}^d.$$

Rearranging (collecting the z and y terms):

$$u_t^{\mathrm{target}}(y \mid z) = \left(\dot{\alpha}_t - \frac{\dot{\beta}_t}{\beta_t}\alpha_t\right) z + \frac{\dot{\beta}_t}{\beta_t} y. \qquad \square$$

The Marginalization Trick

Theorem 4.10 (Marginalization trick)

Let $u_t^{\mathrm{target}}(\cdot \mid z)$ be a conditional vector field satisfying (4.22). Then the marginal vector field

$$u_t^{\mathrm{target}}(x) = \int u_t^{\mathrm{target}}(x \mid z)\, \frac{p_t(x \mid z)\, p_{\mathrm{data}}(z)}{p_t(x)}\, \mathrm{d}z \tag{4.25}$$

yields ODE trajectories that follow the marginal probability path:

$$X_0 \sim p_{\mathrm{init}}, \quad \frac{\mathrm{d}}{\mathrm{d}t} X_t = u_t^{\mathrm{target}}(X_t) \quad \Rightarrow \quad X_t \sim p_t \quad (0 \le t \le 1). \tag{4.26}$$

In particular, $X_1 \sim p_{\mathrm{data}}$.

- Key insight: We can construct the (intractable) marginal vector field from the (tractable) conditional vector field via (4.25)
- The proof uses the continuity equation

The Continuity Equation

Theorem 4.11 (Continuity equation)

Let u_t^{target} be a vector field with $X_0 \sim p_{\text{init}}$. Then $X_t \sim p_t$ for all $0 \le t \le 1$ if and only if

$$\partial_t p_t(x) = -\mathrm{div}(p_t\, u_t^{\text{target}})(x) \quad \textit{for all } x \in \mathbb{R}^d,\ 0 \le t \le 1, \tag{4.27}$$

where $\mathrm{div}(v_t)(x) = \sum_{i=1}^{d} \frac{\partial}{\partial x_i} v_t^i(x)$ is the divergence operator.

- Interpretation: The left side $\partial_t p_t(x)$ measures how probability density changes at x. The right side $-\mathrm{div}(p_t u_t)$ measures the net inflow of probability mass via the vector field. Since mass is conserved, these must be equal
- This is a special case ($\sigma_t = 0$) of the Fokker–Planck equation (Theorem 4.14), whose proof is given later. We use this result to prove the Marginalization Trick

Proof of the Marginalization Trick

We show that $u_t^{\mathrm{target}}(x)$ defined in (4.25) satisfies the continuity equation (4.27).

$$
\begin{aligned}
\partial_t p_t(x) &\overset{(i)}{=} \partial_t \int p_t(x \mid z)\, p_{\mathrm{data}}(z)\, \mathrm{d}z \\
&= \int \partial_t p_t(x \mid z)\, p_{\mathrm{data}}(z)\, \mathrm{d}z \\
&\overset{(ii)}{=} \int -\mathrm{div}\big(p_t(\cdot \mid z)\, u_t^{\mathrm{target}}(\cdot \mid z)\big)(x)\, p_{\mathrm{data}}(z)\, \mathrm{d}z \\
&\overset{(iii)}{=} -\mathrm{div}\left(\int p_t(x \mid z)\, u_t^{\mathrm{target}}(x \mid z)\, p_{\mathrm{data}}(z)\, \mathrm{d}z\right)
\end{aligned}
$$

where:

- (*i*) definition of $p_t(x)$ as $\int p_t(x \mid z)\, p_{\mathrm{data}}(z)\, \mathrm{d}z$ (equation (4.15));
- (*ii*) the conditional vector field satisfies the continuity equation: $\partial_t p_t(\cdot \mid z) = -\mathrm{div}(p_t(\cdot \mid z)\, u_t^{\mathrm{target}}(\cdot \mid z))$ (this holds by hypothesis (4.22) and Theorem 4.11 applied to each fixed z);
- (*iii*) interchange of div_x and $\int \mathrm{d}z$ (justified by Fubini–Tonelli, using the regularity of u_t^{target} and p_t). Passing ∂_t under the integral from line 1 to line 2 is similarly justified by dominated convergence.

Proof of the Marginalization Trick (cont.)

Continuing:

$$
\begin{aligned}
&\overset{(iv)}{=} -\mathrm{div}\left(p_t(x) \int u_t^{\mathrm{target}}(x \mid z)\, \frac{p_t(x \mid z)\, p_{\mathrm{data}}(z)}{p_t(x)}\, \mathrm{d}z\right) \\
&\overset{(v)}{=} -\mathrm{div}\big(p_t\, u_t^{\mathrm{target}}\big)(x),
\end{aligned}
$$

where:

- *(iv)* factor out $p_t(x) > 0$ (the marginal density is strictly positive on its support, since it is a mixture of Gaussians);
- *(v)* recognise the definition of $u_t^{\mathrm{target}}(x)$ in (4.25).

The first and last expressions give $\partial_t p_t(x) = -\mathrm{div}(p_t\, u_t^{\mathrm{target}})(x)$, which is the continuity equation (4.27) for the marginal vector field. By Theorem 4.11, this implies $X_t \sim p_t$ for $0 \leq t \leq 1$. $\square$

The Score Function

Definition 4.12 (Score function)

The *marginal score function* *of* p_t *is*

$$\nabla \log p_t(x) = \frac{\nabla p_t(x)}{p_t(x)}.$$

The *conditional score function* *of* $p_t(\cdot \mid z)$ *is* $\nabla \log p_t(x \mid z)$.

- The marginal score can be expressed via conditional scores:

$$\nabla \log p_t(x) = \int \nabla \log p_t(x \mid z) \, \frac{p_t(x \mid z) \, p_{\mathrm{data}}(z)}{p_t(x)} \, \mathrm{d}z \tag{4.28}$$

- The conditional score is typically known analytically

Score Function for Gaussian Paths

Example 4.13 (Score for Gaussian conditional probability paths)

For $p_t(x \mid z) = \mathcal{N}(x; \alpha_t z, \beta_t^2 I_d)$*, the conditional score is*

$$\nabla \log p_t(x \mid z) = -\frac{x - \alpha_t z}{\beta_t^2}. \tag{4.29}$$

- Proof. The Gaussian density is $\mathcal{N}(x; \mu, \sigma^2 I_d) = (2\pi\sigma^2)^{-d/2} \exp\left(-\frac{\|x-\mu\|^2}{2\sigma^2}\right)$. Taking $\nabla_x \log$ gives $-\frac{x-\mu}{\sigma^2}$. With $\mu = \alpha_t z$ and $\sigma^2 = \beta_t^2$, the result follows
- Key property: The conditional score is *linear* in x. This is a special feature of Gaussian distributions

From ODEs to SDEs: Extending the Continuity Equation

- So far, we have derived the training target for flow models (ODEs):
  - Conditional vector field $u_t^{\mathrm{target}}(x \mid z)$ generates the conditional path $p_t(\cdot \mid z)$
  - Marginalization trick (Theorem 4.10): the marginal vector field generates p_t
  - The proof relied on the continuity equation (4.27)
- Goal: Extend this to diffusion models (SDEs) with $\sigma_t > 0$
- Problem: The continuity equation only governs ODEs. For SDEs, we need to account for the additional spreading of probability mass caused by Brownian noise
- Solution: The Fokker–Planck equation — the SDE analogue of the continuity equation. It adds a second-order (diffusion) term $\frac{\sigma_t^2}{2}\Delta p_t$ to capture the effect of noise
- We will use the score function $\nabla \log p_t$ (just defined) to connect the Fokker–Planck equation back to a concrete SDE formula for diffusion models

The Fokker–Planck Equation

Theorem 4.14 (Fokker–Planck equation)

Let p_t be a probability path and consider the SDE

$$X_0 \sim p_{\mathrm{init}}, \quad \mathrm{d}X_t = u_t(X_t)\,\mathrm{d}t + \sigma_t\,\mathrm{d}W_t.$$

Then X_t has distribution p_t for all $0 \leq t \leq 1$ if and only if the Fokker–Planck equation holds:

$$\partial_t p_t(x) = -\mathrm{div}(p_t u_t)(x) + \frac{\sigma_t^2}{2}\Delta p_t(x) \quad \textit{for all } x \in \mathbb{R}^d,\ 0 \leq t \leq 1, \tag{4.30}$$

where $\Delta w_t(x) = \sum_{i=1}^d \frac{\partial^2}{\partial x_i^2} w_t(x) = \mathrm{div}(\nabla w_t)(x)$ is the Laplacian.

- For $\sigma_t = 0$: reduces to the continuity equation (4.27)
- The additional term $\frac{\sigma_t^2}{2}\Delta p_t$ accounts for the spreading of probability mass due to Brownian noise (cf. the heat equation)

Proof of the Fokker–Planck Equation (Necessity)

Strategy: Let $f \in C_c^\infty(\mathbb{R}^d)$ (smooth, compactly supported). We show: if $X_t \sim p_t$ then (4.30) holds in the sense of distributions.

Step 1 (Itô's formula). Apply Itô's formula to $f(X_t)$:

$$\begin{aligned}
\mathrm{d}f(X_t) &= \nabla f(X_t)^T \mathrm{d}X_t + \frac{1}{2}\mathrm{tr}\big(\sigma_t^2 \nabla^2 f(X_t)\big)\,\mathrm{d}t \\
&= \nabla f(X_t)^T u_t(X_t)\,\mathrm{d}t + \sigma_t \nabla f(X_t)^T \mathrm{d}W_t + \frac{\sigma_t^2}{2}\Delta f(X_t)\,\mathrm{d}t.
\end{aligned}$$

Here we used $\mathrm{tr}(\nabla^2 f) = \Delta f$ and the Itô correction rule $\mathrm{d}W_t^i\,\mathrm{d}W_t^j = \delta_{ij}\,\mathrm{d}t$.

Proof of the Fokker–Planck Equation (Necessity, cont.)

Step 2 (Integrate and take expectation). Integrate from t to $t+h$ and take expectation. The stochastic integral $\int_t^{t+h} \sigma_s \nabla f(X_s)^T \mathrm{d}W_s$ is a martingale (since $f \in C_c^\infty$ implies ∇f is bounded), so its expectation vanishes:

$$\mathbb{E}[f(X_{t+h})] - \mathbb{E}[f(X_t)] = \int_t^{t+h} \mathbb{E}\left[\nabla f(X_s)^T u_s(X_s) + \frac{\sigma_s^2}{2} \Delta f(X_s)\right] \mathrm{d}s.$$

Dividing by h and taking $h \to 0$ (by the Lebesgue differentiation theorem):

$$\frac{\mathrm{d}}{\mathrm{d}t} \mathbb{E}[f(X_t)] = \mathbb{E}\left[\nabla f(X_t)^T u_t(X_t) + \frac{\sigma_t^2}{2} \Delta f(X_t)\right].$$

Proof of the Fokker–Planck Equation (Necessity, cont.)

Step 3 (Write as integrals). Using $X_t \sim p_t$:

$$\frac{\mathrm{d}}{\mathrm{d}t}\int_{\mathbb{R}^d} f(x)\, p_t(x)\,\mathrm{d}x = \int_{\mathbb{R}^d}\left[\nabla f(x)^T u_t(x) + \frac{\sigma_t^2}{2}\Delta f(x)\right] p_t(x)\,\mathrm{d}x.$$

Pass $\frac{\mathrm{d}}{\mathrm{d}t}$ under the integral on the LHS (justified by dominated convergence: f has compact support and $t \mapsto p_t$ is smooth):

$$\int_{\mathbb{R}^d} f(x)\,\partial_t p_t(x)\,\mathrm{d}x = \int_{\mathbb{R}^d} \nabla f(x)^T \big(p_t(x)\,u_t(x)\big)\,\mathrm{d}x + \int_{\mathbb{R}^d} \frac{\sigma_t^2}{2}\Delta f(x)\,p_t(x)\,\mathrm{d}x.$$

Proof of the Fokker–Planck Equation (Necessity, cont.)

Step 4 (Integration by parts). Since $f \in C_c^\infty$, all boundary terms vanish:

- $\int_{\mathbb{R}^d} \nabla f^T (p_t u_t)\,\mathrm{d}x = -\int_{\mathbb{R}^d} f\,\mathrm{div}(p_t u_t)\,\mathrm{d}x$ (divergence theorem)
- $\int_{\mathbb{R}^d} \Delta f \cdot p_t\,\mathrm{d}x = \int_{\mathbb{R}^d} f \cdot \Delta p_t\,\mathrm{d}x$ (Green's second identity, applied twice)

Substituting:

$$\int_{\mathbb{R}^d} f(x)\,\partial_t p_t(x)\,\mathrm{d}x = \int_{\mathbb{R}^d} f(x)\left[-\mathrm{div}(p_t u_t)(x) + \frac{\sigma_t^2}{2}\Delta p_t(x)\right]\mathrm{d}x.$$

This holds for all $f \in C_c^\infty(\mathbb{R}^d)$. By the fundamental lemma of the calculus of variations (du Bois-Reymond):

$$\partial_t p_t(x) = -\mathrm{div}(p_t u_t)(x) + \frac{\sigma_t^2}{2}\Delta p_t(x). \qquad \square$$

Proof of the Fokker–Planck Equation (Sufficiency)

Claim: If p_t satisfies (4.30) with $p_0 = p_{\mathrm{init}}$, then $X_t \sim p_t$ for all $t \in [0,1]$.

Proof.

- Let q_t denote the law of X_t. By necessity (just proved), q_t satisfies:

$$\partial_t q_t = -\mathrm{div}(q_t u_t) + \frac{\sigma_t^2}{2}\Delta q_t, \qquad q_0 = p_{\mathrm{init}}.$$

- By hypothesis, p_t satisfies the same PDE with the same initial condition $p_0 = p_{\mathrm{init}}$
- The Fokker–Planck equation is a *linear* parabolic PDE of the form $\partial_t \rho = \mathcal{L}^* \rho$, where $\mathcal{L}^*$ is the formal L^2-adjoint of the infinitesimal generator $\mathcal{L}f = u_t^T \nabla f + \frac{\sigma_t^2}{2}\Delta f$
- Under our regularity assumptions (u_t Lipschitz, σ_t continuous), uniqueness of distributional solutions in the class of probability densities holds[15]
- Therefore $p_t = q_t$ for all $t \in [0,1]$. □

[15] Le Bris, C., Lions, P.-L. (2008). Existence and uniqueness of solutions to Fokker–Planck type equations with irregular coefficients. *Comm. Partial Differential Equations*, 33(7), 1272–1317.

The SDE Extension Trick

Theorem 4.15 (SDE extension trick)

Define the conditional and marginal vector fields $u_t^{\text{target}}(x \mid z)$ *and* $u_t^{\text{target}}(x)$ *as before. Then for any diffusion coefficient* $\sigma_t \geq 0$*, the SDE*

$$X_0 \sim p_{\text{init}}, \quad \mathrm{d}X_t = \left[u_t^{\text{target}}(X_t) + \frac{\sigma_t^2}{2} \nabla \log p_t(X_t) \right] \mathrm{d}t + \sigma_t \, \mathrm{d}W_t \tag{4.31}$$

satisfies $X_t \sim p_t$ *for* $0 \leq t \leq 1$*. In particular,* $X_1 \sim p_{\text{data}}$*.*

- For $\sigma_t = 0$: recovers the ODE $\mathrm{d}X_t = u_t^{\text{target}}(X_t)\,\mathrm{d}t$
- The same identity holds replacing marginal quantities by conditional ones
- Key insight: Adding noise ($\sigma_t > 0$) requires a correction term $\frac{\sigma_t^2}{2} \nabla \log p_t$ to maintain the same marginal path

Proof of the SDE Extension Trick

We verify the Fokker–Planck equation (4.30) for the SDE (4.31):

$$\begin{aligned}
\partial_t p_t(x) &\overset{(i)}{=} -\mathrm{div}(p_t u_t^{\text{target}})(x) \\
&\overset{(ii)}{=} -\mathrm{div}(p_t u_t^{\text{target}})(x) - \frac{\sigma_t^2}{2}\Delta p_t(x) + \frac{\sigma_t^2}{2}\Delta p_t(x) \\
&\overset{(iii)}{=} -\mathrm{div}(p_t u_t^{\text{target}})(x) - \mathrm{div}\left(\frac{\sigma_t^2}{2}\nabla p_t\right)(x) + \frac{\sigma_t^2}{2}\Delta p_t(x)
\end{aligned}$$

$$\begin{aligned}
&\overset{(iv)}{=} -\mathrm{div}(p_t u_t^{\text{target}})(x) - \mathrm{div}\left(p_t \cdot \frac{\sigma_t^2}{2}\nabla \log p_t\right)(x) + \frac{\sigma_t^2}{2}\Delta p_t(x) \\
&\overset{(v)}{=} -\mathrm{div}\left(p_t\left[u_t^{\text{target}} + \frac{\sigma_t^2}{2}\nabla \log p_t\right]\right)(x) + \frac{\sigma_t^2}{2}\Delta p_t(x),
\end{aligned}$$

where:

- (*i*) By the Marginalization Trick (Theorem 4.10), u_t^{target} satisfies the continuity equation $\partial_t p_t = -\mathrm{div}(p_t u_t^{\text{target}})$;
- (*ii*) add and subtract $\frac{\sigma_t^2}{2}\Delta p_t$;
- (*iii*) $\Delta p_t = \mathrm{div}(\nabla p_t)$ (definition of the Laplacian);
- (*iv*) $\nabla p_t = p_t \nabla \log p_t$ (valid where $p_t > 0$; holds on the support since p_t is a Gaussian mixture);

Langevin Dynamics

Remark 4.16 (Langevin dynamics)

A famous special case: if the probability path is static ($p_t = p$ for all t), set $u_t^{\text{target}} = 0$ to obtain:

$$\mathrm{d}X_t = \frac{\sigma_t^2}{2} \nabla \log p(X_t)\, \mathrm{d}t + \sigma_t\, \mathrm{d}W_t. \tag{4.32}$$

- Since $p_t = p$ is static, $\partial_t p_t = 0$, and p is a *stationary distribution* of the SDE
- If $X_0 \sim p$, then $X_t \sim p$ for all $t \geq 0$
- Even if $X_0 \not\sim p$, under mild conditions $X_t \to p$ as $t \to \infty$ (convergence to equilibrium)
- Langevin dynamics is the foundation of many MCMC methods in Bayesian statistics, molecular dynamics, and sampling algorithms

Summary: Constructing the Training Target

Summary

1. Choose a conditional probability path $p_t(x \mid z)$ with $p_0(\cdot \mid z) = p_{\mathrm{init}}$, $p_1(\cdot \mid z) = \delta_z$
2. Find a conditional vector field $u_t^{\mathrm{target}}(x \mid z)$ whose ODE follows $p_t(\cdot \mid z)$
3. The marginal vector field $u_t^{\mathrm{target}}(x) = \int u_t^{\mathrm{target}}(x \mid z) \frac{p_t(x|z) p_{\mathrm{data}}(z)}{p_t(x)} \, \mathrm{d}z$ converts p_{init} into p_{data}
4. Extending to SDEs: $\mathrm{d}X_t = [u_t^{\mathrm{target}}(X_t) + \frac{\sigma_t^2}{2} \nabla \log p_t(X_t)] \, \mathrm{d}t + \sigma_t \, \mathrm{d}W_t$

For Gaussian probability paths $p_t(\cdot \mid z) = \mathcal{N}(\alpha_t z, \beta_t^2 I_d)$:

$$u_t^{\mathrm{target}}(x \mid z) = \left(\dot{\alpha}_t - \frac{\dot{\beta}_t}{\beta_t} \alpha_t \right) z + \frac{\dot{\beta}_t}{\beta_t} x, \qquad \nabla \log p_t(x \mid z) = -\frac{x - \alpha_t z}{\beta_t^2}.$$

End of Week 4